

The Straight-Line Book

Two-Ledger Law $\Delta \mu = -\ln J_{\text{act}}$ over the Whole Frontier: slope -1, and the Shaping as a Monotone

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L.O.R.E. Framework (Law of Repulsive Emanation)

Abstract

Ch.82 found the two ledgers - the tilt-center translation $\Delta\mu$ and the exponential information book $\ln J_{\text{act}}$ - agreed to $\sim 3\%$ on one mild coin. This chapter measures the FULL engaged frontier and promotes that agreement into a LAW: $\Delta\mu = -\ln J_{\text{act}}$ holds EXACTLY in the mild-coin (constant-detuning) limit, with a through-origin slope of $-1.012 \sim -1$ and a mean deviation $|\text{dev}| = 0.0006$; and the deviation is a MONOTONE function of the leverage ($0.003 \rightarrow 0.024$, verified monotone), which is precisely the Ch.81 shaping, now traced as a curve against the coin rather than at single points. At the control end both $\Delta\mu$ and $\ln J_{\text{act}}$ vanish together, so the slope is a $0/0$ whose removable value is -1 - the mirror's constant, the same removable value as $P'(0)/P(0) = 1/2$ and the tilt center $a(1/2) = 0$.

1. Why a Law

Ch.82 compared two books at four leverages. A single coincidence could be luck; a LAW is a straight line through the origin with a determined slope that the whole frontier follows, and whose residual is itself meaningful. On the same round-trip trap ($\lambda 1 \rightarrow 2 \rightarrow 1$, $\Delta\mu_F = 0$, $D = \beta = 1$, Heun SRK2, seed 42), sweep the fast/slow leverage of the median-bit demon ($0.5/0.5$ null, $0.35/2$, $0.25/4$, $0.15/6$, $0.10/8$, $0.05/16$):

mid coin ($0.35/2$):	$\Delta\mu = -0.0701$, $-\ln J = -0.0726$, $\text{dev} +0.0026$
$0.25/4$:	-0.0979 vs -0.0941 , $\text{dev} -0.0038$
$0.15/6$:	-0.1231 vs -0.1105 , $\text{dev} -0.0126$
$0.10/8$:	-0.1368 vs -0.1197 , $\text{dev} -0.0171$
$0.05/16$ (harvest):	-0.1513 vs -0.1269 , $\text{dev} -0.0244$

2. Slope -1 in the Mild Limit

The through-origin slope over the two mildest engaged rows is -1.012 (expected -1), mean $|\text{dev}| = 0.0006$: a weak coin is entirely a constant detuning, so its detuning dollar-for-dollar equals its information book. The mild law is exact. A strong coin keeps the same lead but adds a deviation, and that deviation RISES monotonically with the leverage - verified monotone. The free fit over the whole frontier (slope -0.85 , intercept $+0.005$) is the strong rows bending the line; the slope -1 belongs to the mild limit.

3. The Deviation Is the Shaping Curve

The per-row $|\text{deviation}| = |\Delta\mu + \ln J_{\text{act}}|$ is nothing other than the Ch.81 state-dependence (netR), now measured as a function of the coin:

leverage	$ \text{dev} $ (shaping)
$0.35/2$	0.0026
$0.25/4$	0.0038
$0.15/6$	0.0126
$0.10/8$	0.0171
$0.05/16$	0.0244 (monotone)

The two-ledger law is exact to the extent the demon is a constant; the deviation is the conditional part that grows as the bit is engineered harder. Both are measured; neither is assumed.

4. The $0/0$ That Fixes the Slope

At the null/control end (no feedback) $\Delta\mu \rightarrow 0$ and $\ln J_{\text{act}} \rightarrow 0$ TOGETHER - the detuning and the information book both vanish at reversibility - so the slope $\Delta\mu / \ln J_{\text{act}}$ is a $0/0$. Its removable value is the measured -1 : the same constant the mirror carried at $P'(0)/P(0) = 1/2$ and the tilt center $a(1/2) = 0$ (Ch.77, Ch.80). One removable value, now dressed as the slope of a law.

Main Results

THEOREM

The Two-Ledger Law: on the feedback round trip $\Delta\mu = -\ln J_{\text{act}}$ holds EXACTLY in the mild-coin limit (through-origin slope $-1.012 \sim -1$, mean $|\text{dev}| = 0.0006$), and the deviation is a MONOTONE function of the leverage ($0.003 \rightarrow 0.024$), the Ch.81 shaping as a curve.

COROLLARY

The Slope Is the Mirror's Constant: at the control end both $\Delta\mu$ and $\ln J_{\text{act}}$ vanish, so the slope is $0/0$ whose removable value is -1 - the same removable value as $P'(0)/P(0) = 1/2$ and the tilt center $a(1/2) = 0$ (Ch.77, Ch.80).

COROLLARY

The Shaping Is Monotone: a pure detuning has zero deviation; the stronger the state-dependence of the bit, the larger $|\Delta\mu + \ln J_{\text{act}}|$ - one coin, two ledgers, a straight line in the weak limit and a measurable curvature in the strong.

Conclusion

The two books of Ch.74/75 and Ch.80/81, shown agreeing on one coin in Ch.82, are now a law: $\Delta\mu = -\ln J_{\text{act}}$, exact for the

constant detuning that is a weak demon and bent by exactly the shaping that a strong demon adds, monotonically. The 0/0 at the control end fixes the slope at -1, the mirror's own constant. Eight lenses on one equality and one law: the demon's price is a straight line through the origin.

References

- [1] Sagawa, T. & Ueda, M. (2010). Phys. Rev. Lett. 104, 090602.
- [2] Crooks, G.E. (1999). Phys. Rev. E 60, 2721 (detailed theorem).
- [3] Parrondo, J.M.R., Horowitz, J.M., Sagawa, T. (2015). Nat. Phys. 11, 131.
- [4] Szilard, L. (1929); Landauer, R. (1961) (bit face value, erasure).
- [5] Book chain: Ch.74-75 (demon bill), Ch.80 (tilt center), Ch.81 (detuning-shaping split), Ch.82 (two ledgers one price).